AV AI VELOCITY ARCHITECT INTELLIGENCE · ISSUE 02

WEEK OF 5–11 SEPTEMBER 2026

Who Is Allowed To Say No

A clock, a server-side policy and a token audience can each answer before the clinician does. The audit trail followed none of them.

16	2	8	5
packages swept	mechanisms executed	papers read	models moved

Vijay Zharotia
Healthcare AI architecture

WHY AI VELOCITY

Agents that act inherit obligations.

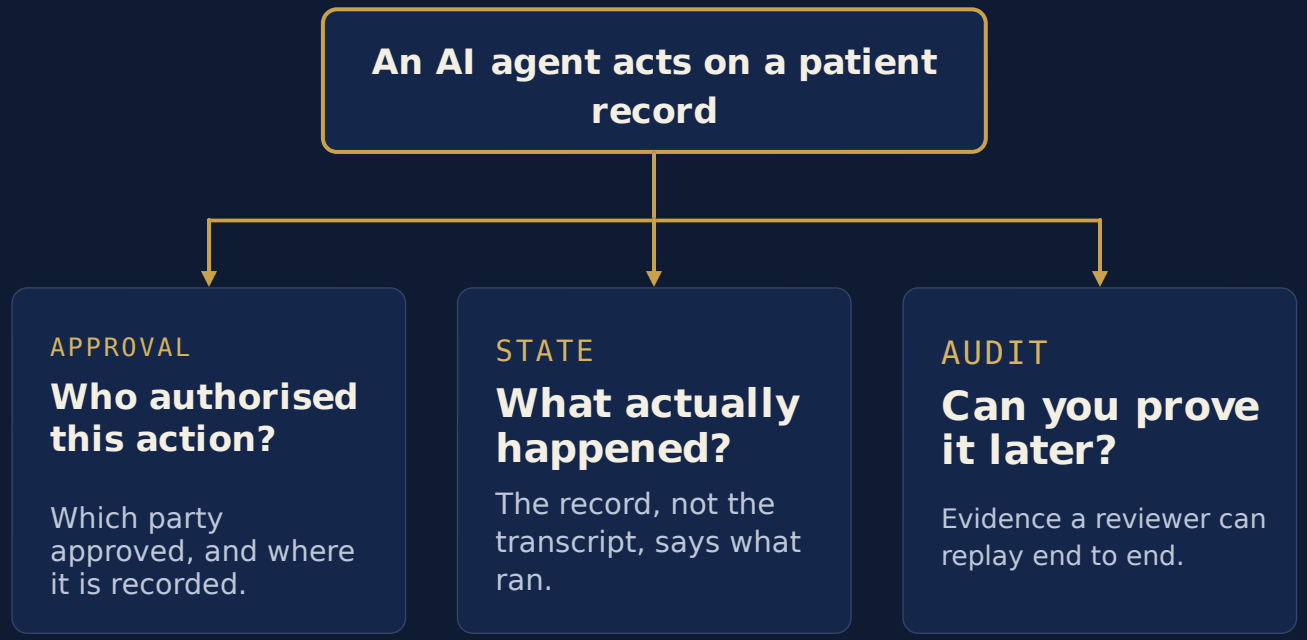
A weekly architecture brief for teams putting AI agents into healthcare workflows. We read the releases, run the mechanism, and say what to change.

FOR ARCHITECTS

The mechanism, the code, the move.

FOR PRODUCT MANAGERS

The decision, the risk, the date.



EVERY CLAIM CARRIES ITS EVIDENCE

EXECUTED

We ran it. The numbers are observed output.

DOCS

Read from documentation or source.

UNVERIFIED

Not confirmed. Vendor numbers carry an asterisk.

THE WEEK IN ONE VIEW

From where the clinician signs to who answers first

CONTEXT

Last week: the signature moved

Issue 01 found four new places a clinician can sign. This week the frameworks decide how long that signature may take and which server it opens.

→

PROBLEM

Something answers before the clinician

A clock, a server-side policy and a token audience can each decide first, and none of them wrote anything to the transcript.

→

SOLUTION

Record every answer, human or not

Persist what decided, check the approval budget before asking for a signature, and bind every token to one server.

TOKENS · 04–05
One token per server

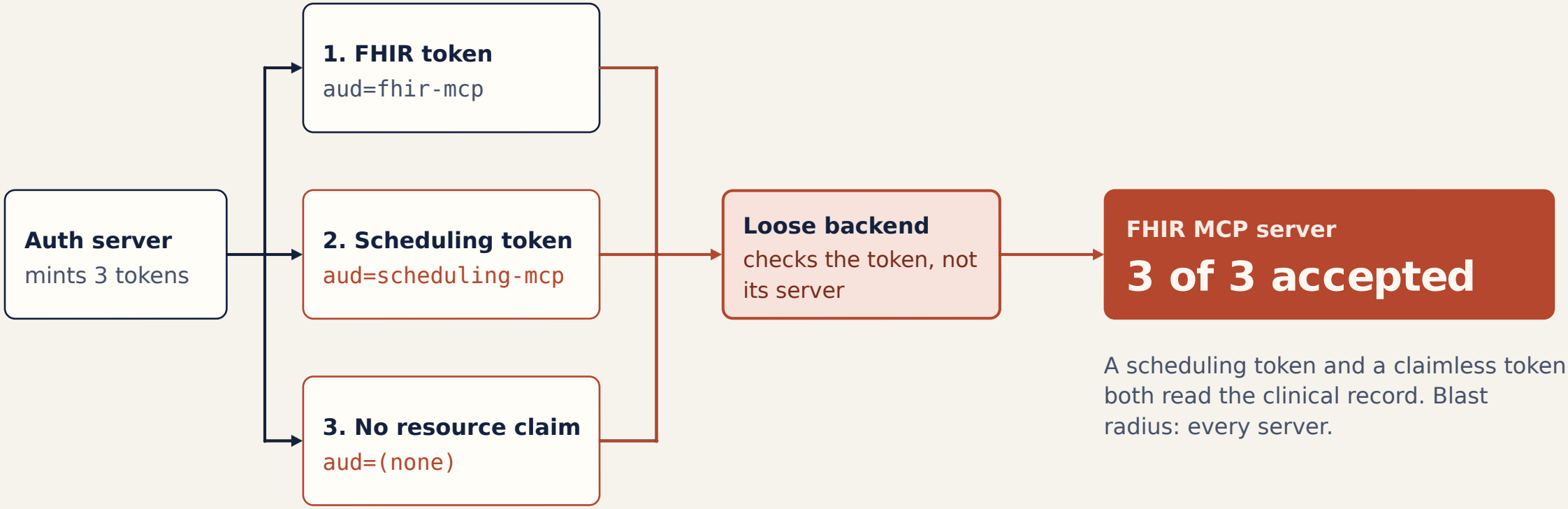
APPROVAL · 06
Who answers first

STATE · 07
Resume and drift

EVIDENCE · 08
What the model believes

COST, DATES · 09–10
Price anchor, dates

One token opened every server



A scheduling token and a claimless token both read the clinical record. Blast radius: every server.

EXECUTED MECHANISM · SOLUTION · MCP PYTHON SDK 2.2.0

Pin each token to one server

STEP 1

Bind tokens to a resource

The auth server sets an RFC 8707 resource claim on every token.

→

STEP 2

Pin the backend

Set resource_server_url on BearerAuthBackend.

→

STEP 3

Alert on the warning

Roll out with validation off and watch who would be refused.

→

STEP 4

Then go strict

Flip once every token carries its server.

3 of 3

tokens accepted by the loose backend

1 of 3

accepted once pinned: the FHIR token

2 refused

wrong resource, and no resource at all

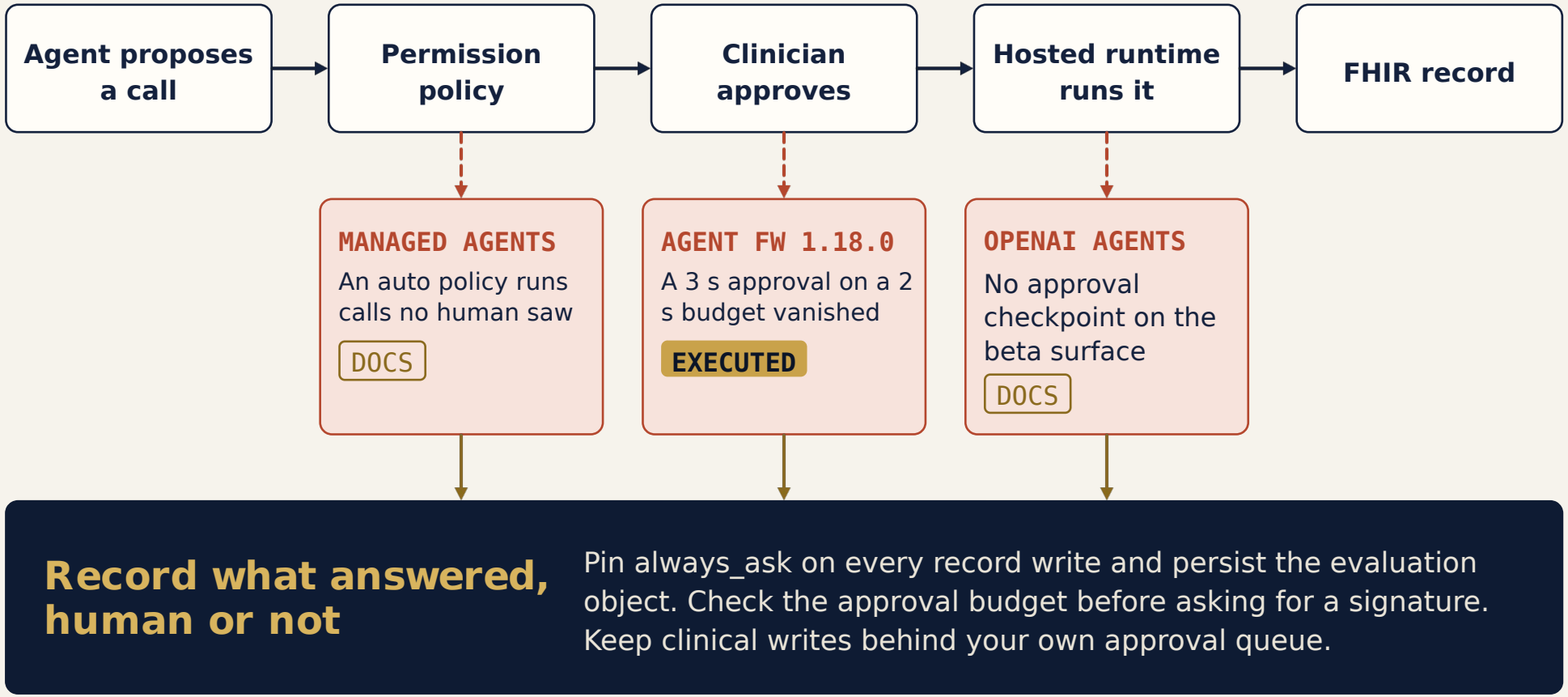
> A refusal is the cheapest audit event there is: no model, no prompt, no clinician. test_mcp_token_audience.py reproduces the table in under a minute.

Three things answered before the clinician

CONTEXT
how it runs

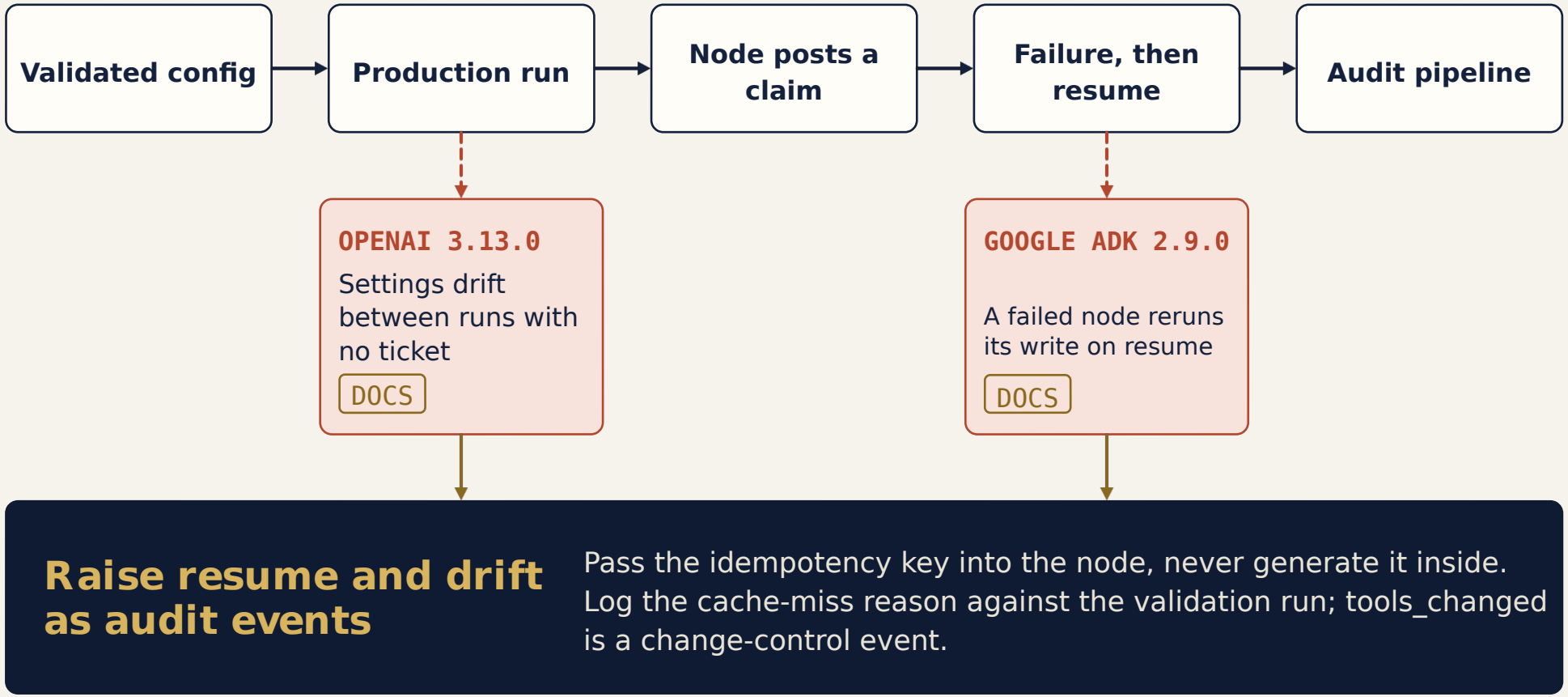
PROBLEM
where it broke

SOLUTION
what to build



Two audit events your pipeline does not raise

CONTEXT
how it runs



PROBLEM
where it broke

SOLUTION
what to build

Raise resume and drift as audit events

Pass the idempotency key into the node, never generate it inside. Log the cache-miss reason against the validation run; tools_changed is a change-control event.

The model believes what it is handed

CONTEXT
how it runs

Agent gets a task → **Reads the full chart** → **Session is compacted** → **A tool returns a value** → **Acts on the record**

PROBLEM
where it broke

OBGYNLONGBENCH
10–23 pts
lost when models must find the evidence

CONSTRAINT LOSS
87%
loss of control when compaction drops the rules

UNRELIABLE TOOLS
5.3%
of the time agents warned the user

SOLUTION
what to build

Put verification outside the model

Name the decision target before retrieving, pin authorisation rules as an uncompressible block, and check tool outputs with ranges and independent recomputation.

COST · LIST PRICES ON 11 SEPTEMBER

The price anchor moved to an open licence

MODEL	COST PER 10,000 DOCS	INDEX
DeepSeek V4.1 Flash	**$18; $9 off-peak**	**40**
Gemini 3.8 Flash	**$48.75**	**41**
Claude Opus 5	**$325**	**51**
Claude Fable 5.1	**$650**	**53**
GPT-6 Astra	**$650**	**53**

4,000 in and 500 out tokens per document. Per M tokens in/out: DeepSeek $0.30/$1.20 (half off-peak), Gemini 3.8 Flash $0.75/$3.75, Opus 5 $5/$25, Fable 5.1 and Astra $10/$50. Index: Artificial Analysis v4.3, rebased 7 Sep; not comparable with Issue 01.

CONTEXT

Bounded extraction was priced against closed Flash-tier models.

SHIFT

MIT weights now do it for $18, one index point under Gemini 3.8 Flash.

DECISION

Host the 552B MoE behind your boundary, or accept the API’s jurisdiction. Opus 5 stays the reasoning default at half the flagship price.

Dates around this week



● REGULATORY AND FUNDING ● INDEX ● MODEL

If you ship SaMD on a foundation model, comment.

Identity, durability and isolation, not wrappers

LANGCHAIN

Fork and isolated subagents

Plus per-caller OAuth. Fork for workers, isolated for verifiers.

PYDANTIC

Durable steps on AWS Lambda

Replayed steps with no re-issued writes for pydantic-ai.

OPENAI

Self-hosted sandboxes

For PHI-bearing code execution. The Agents API beta is US-only, no zero retention.

ANTHROPIC

July’s escapes traced

An alignment assessment names an eval sandbox on the open internet as root cause.

NVIDIA

BioNeMo Inference Runtime

Plus encode-prefill-decode disaggregation for serving.

QUIET THIS WEEK

LangGraph, LlamaIndex core, Semantic Kernel, the MCP spec, HL7 publications, open.epic.

Seven moves, with what forces each

Move	Why	By
Pin always_ask on every record write; persist the evaluation	auto is not a human checkpoint	This sprint
Check the approval budget before asking for a signature	Late approvals vanish	This sprint
Pin MCP backends to their server URL; alert on the warning	One token, every server	Next rollout
Pass idempotency keys into ADK nodes	Failed nodes rerun writes	Before ADK 2.9.0
Log cache-miss reasons against the validation run	Drift with no ticket	Next release
Re-run any model decision taken on Issue 01 scores	Index rebased to v4.3	Now
File an FDA comment if you ship SaMD on a foundation model	Docket FDA-2026-N-7874	19 Oct

AV AI VELOCITY

THIS WEEK’S POSITION

A refusal is the cheapest audit event there is.

Long-form edition
Every mechanism, source and date

Lab package
Runs clean, no API key needed

New issue every Friday
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github.com/zkumar/ai-velocity-weekly